

KALEB RODRIGUEZ

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EDUCATION

Middleton High School — CSIT Magnet

Tampa, FL

High School Student | GPA: 4.4/5.0 (Weighted) | Expected Graduation: 2027

AP Coursework: AP Biology, AP Computer Science A, AP Computer Science Principles, AP Calculus AB, AP Calculus BC, AP Physics 1, AP Seminar, AP Microeconomics, AP Precalculus, AP Art History

University of South Florida — Dual Enrollment Student, Ongoing | GPA: 4.0

Tampa, FL

University of Florida — Dual Enrollment Student, Ongoing

Gainesville, FL

RESEARCH EXPERIENCE

MIT CSAIL — AI Mantis Rising Scholars Program

Remote

Research Scholar

May 2026 – Present

- Remote research program under Prof. Manolis Kellis (MIT CSAIL/EECS); contribute to the Mantis Visual Data Science Platform for AI-driven analysis of medical, biological, and scientific datasets
- Develop AI and data science tools spanning full-stack engineering, ML pipelines, data visualization, and agent-based systems, applying industry-standard workflows including version control, agile development, and collaborative software deployment

USF College of Engineering — Koria Lab

Tampa, FL

Laboratory Research Assistant

June 2026 – Present

- Conduct wet lab research on elastin-like polypeptide (ELP) fusion proteins for drug delivery and regenerative medicine applications
- Perform bacterial protein expression, cell lysis via sonication, and protein purification using inverse transition cycling (ITC)

USF Computational Biology Research

Tampa, FL

Research Assistant

January 2026 – Present (on hiatus)

- Apply ESM2-8M protein language model embeddings to predict hemolytic peptide activity as a binary classification task

Oakland University — NLP Research Lab

Remote

Research Assistant

April 2026 – Present

- Conduct NLP research on health misinformation detection and propagation on social media, building classification models that leverage prior biomedical ML experience to flag unreliable health claims

NASA GeneLab

Remote

Student Research Intern

June 2026 – August 2026

- Selected for a competitive bioinformatics and space biology program; analyze omics datasets from spaceflight experiments to study biological responses to spaceflight conditions

ThinkNeuro, LLC

Remote

Co-Associate

June 2026 – August 2026

- Help lead and coordinate a student neuroscience research team investigating neurodegenerative diseases through literature review, bibliometric analysis, and scientific manuscript development

Research Intern

January 2026 – April 2026

- Investigated operational similarities between ML models and dopaminergic/cholinergic neuronal mechanisms; used R for bibliometric analysis; presented a poster and oral defense at the ThinkNeuro Spring Research Symposium

PROFESSIONAL EXPERIENCE

ThinkNeuro, LLC

Remote

Software Engineering Intern

January 2026 – April 2026

- Designed and built MedAlert, a healthcare iOS app in Swift/SwiftUI with scheduled dosing, notification alerts, medication tracking, and user-centered UI/UX

UGC Co-Lead

January 2026 – August 2026

- Oversee production of short-form digital content driving ThinkNeuro's digital marketing presence, reaching 100K+ views

UOTY — Faith-Based Organization

Tampa, FL

Founding Member & Head of Public Relations

July 2020 – Present

- Co-founded the PR committee; design digital media, coordinate speakers, and create social content to strengthen youth community presence

Back on Track America — Middleton Chapter

Tampa, FL

Founder & Chapter Lead

December 2025 – Present

- Founded and lead the school's Back on Track chapter, addressing food insecurity and homelessness; led food distribution events with Metropolitan Ministries and conducted nonprofit outreach for student volunteers

PROJECTS

Posture+ — AI Posture Coach

Co-Founder & Developer

February 2026 – May 2026

- Built a real-time AI posture monitoring app for TYE Tampa Bay using MediaPipe Pose to detect slouching, uneven shoulders, and head tilt with a live 0–100 score; designed a SaaS model with free/premium tiers targeting students and remote workers

MedAlert — iOS Application

Developer

January 2026 – April 2026

- Built an iOS medication reminder app in Swift/SwiftUI with scheduled dosing, notification alerts, and medication tracking, with a clean UI/UX designed for real-world patient use

ClearCredit — Degree Audit Tool

clearcredit.vercel.app

Developer

2026 – Present

- Built a full-stack degree audit web app (React, Supabase, Vite) for Florida dual enrollment students that maps HCC, USF, and AP credits against all 21 HCC AA/AS degree pathways in real time
- Built a 'Pick My Best Fit' algorithm that scores degree programs against a student's existing credits; awaiting launch at Middleton High School with planned expansion across Hillsborough County Public Schools

Cramb — Spaced-Repetition Flashcard Extension

kalebrodriguez.github.io/cramb-extension

Developer

June 2026 – Present

- Built a privacy-first, open-source browser extension that automatically converts online reading and viewing into spaced-repetition flashcards to improve long-term retention; submission to the Chrome Web Store is pending review

PUBLICATIONS & PRESENTATIONS

Kingsley, C., Ahad, N., Rodriguez, K., Xiang, A.Q. (2026). Nanoparticle-Mediated Modulation of the PINK1/Parkin Mitophagy Pathway in Parkinson's Disease: A Scoping Review. Research Square (preprint); to be presented at the NRCP National Research Conference, University of Pennsylvania. doi.org/10.21203/rs.3.rs-10308858/v1

Rodriguez, K. (2026). From Neurotransmitters to Algorithms: Evaluating Operational Parallels Between Dopaminergic and Acetylcholinergic Mechanisms and Biologically-Informed Machine Learning Models. Presented at the 2026 NYC Neuromodulation Conference; currently pending publication. doi.org/10.6084/m9.figshare.33115934

Rodriguez, K. (2026). Tracking the Biomarker Landscape: Bibliometric Dynamics in CJD and Rapidly Progressive Dementia Differential Diagnosis. Poster presented at the ThinkNeuro Summer Research Symposium. doi.org/10.6084/m9.figshare.33457435

CERTIFICATIONS

Java Coding Specialist (2026), Python Coding Specialist (2024), CompTIA ITF+ (2024), ITS Networking (2024), ITS Device Configuration & Management (2024), PMI Project Management Ready (2024), Adult First Aid/CPR/AED — Red Cross (2026)

HONORS & AWARDS

AP Scholar with Honor (2025) • TSA Biotechnology Design State Finalist — 7th Place (2026) • FBLA UX Design 4th Place (2026) • FBLA Parliamentary Procedures 2nd Place (2026) • HOSA CPR/First Aid 1st Place (2026) • FBLA State Conference (3x) • Varsity Soccer Letter • Honor Roll • 400+ Service Hours